MATH 415 - FALL 2020 - EXTRA PRACTICE PROBLEMS - WEEK 5

1. *Determine which of the following sets are subspaces of the indicated vector spaces and* **explain why**. *Also sketch the sets W_3 and W_4 and give geometric reasons why W_3 and W_4 either are or are not subspaces of $\mathbb{R}^2$.*

(a) $W_1 = \left\{ \begin{bmatrix} a \\ b \\ c \end{bmatrix} \ : \ a - 2b = c, 4a + 2c = 0 \right\} \subseteq \mathbb{R}^3.$

W_1 is a subspace of $\mathbb{R}^3$, since it is the Nullspace of the matrix

$$\begin{bmatrix} 1 & -2 & -1 \\ 4 & 0 & 2 \end{bmatrix}$$

(b) $W_2 = \left\{ \begin{bmatrix} a-b \\ c \\ a+c \\ a-2b-c \end{bmatrix} \ : \ a, b, c \in \mathbb{R} \right\} \subseteq \mathbb{R}^4.$

Since

$$\left\{ \begin{bmatrix} a-b \\ c \\ a+c \\ a-2b-c \end{bmatrix} \ : \ a, b, c \in \mathbb{R} \right\} = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 0 \\ -2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ -1 \end{bmatrix} \right\}$$

and every span of a set of vectors in $\mathbb{R}^4$ is a subspace of $\mathbb{R}^4$. We can also show this by seeing that W_2 is the column space of

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & -2 & -1 \end{bmatrix}$$

(c) $W_3 = \left\{ \begin{bmatrix} a \\ b \end{bmatrix} \ : \ a \cdot b \geq 0 \right\} \subseteq \mathbb{R}^2.$

W_3 is not a subspace of $\mathbb{R}^2$. Consider the two vectors $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ -1 \end{bmatrix}$. Both are in W_3, because $1 \cdot 0 = 0 \cdot (-1) \geq 0$. But

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

and $1 \cdot (-1) = -1 < 0$. Hence $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ is not in W_3. Hence W_3 is not closed under addition. A sketch of W_3 in $\mathbb{R}^2$ will contain the x-axis, the y-axis and the first and third quadrants. Plotting the above two vectors and their sum on your sketch will show visually how vector addition will leave W_3. Furthermore, the only subspaces of $\mathbb{R}^2$ are (1) the origin, (2) a single line through the origin or (3) the entire plane; its clear from the sketch that W_3 is none of these. [Note: on a quiz or exam, simply sketching W_3 is **not** sufficient justification for why it is not a subspace of $\mathbb{R}^2$, you must give an **explicit** counterexample and state which rule for subspaces is violated (closure under addition in the case of W_3). However, it may be a good idea to use a sketch to help find a counterexample.]

(d) $W_4 = \left\{ \begin{bmatrix} a \\ b \end{bmatrix} \ : \ b \geq 0 \right\} \subseteq \mathbb{R}^2$.

This set is not a subspace of $\mathbb{R}^2$. Consider the vector $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Since $1 \geq 0$, we have that $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ is in W_4. However, $(-1)\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$ is not in W_4, since $-1 < 0$. Hence W_4 is not closed under scalar multiplication. A sketch of W_4 is above x-axis. By plotting our counterexample, we can see visually that W_4 is not closed under scalar multiplication. (Exercise: is W_4 closed under addition?)

(e) $W_5 = \left\{ \begin{bmatrix} a \\ b \\ c \end{bmatrix} \ : \ a - 2b = 1, 4a + 2c = 0 \right\} \subseteq \mathbb{R}^3$.

The zero vector $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ does not satisfy $0 - 2(0) = 1$. Therefore W_5 is not a subspace of $\mathbb{R}^3$.

2. *Are the following sets subspaces of the vector space of all 2×2 matrices $M_{2\times 2}$?*

(a) $\{\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2\times 2} \ : \ ad - bc = 0\}$.

Let S be $\{\begin{bmatrix} a & b \\ c & d \end{bmatrix} \ : \ ad - bc = 0\}$. Then S is not a subspace of $M_{2\times 2}$ since $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \in S$ and $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \in S$, but $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \notin S$. Thus S is not closed under addition.

(b) $\{B \in M_{2\times 2} \ : \ B^T = B\}$. (This is the set of symmetric 2×2 matrices.)

Let V be $\{B \in M_{2\times 2} \ : \ B^T = B\}$. Yes, V is a subspace of $M_{2\times 2}$. We have to check that it contains $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ and is closed under addition and scalar multiplication. First note that $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ is in V, because it is obviously symmetric.

Now take two matrices A, B in V. So we have $A^T = A$ and $B^T = B$. Then we have

$$(A+B)^T = A^T + B^T = A + B.$$

Hence $A + B$ is in V. So V is closed under addition.

Now take a matrix A in V and a scalar r. Since A is in V, we have $A^T = A$. Then we have

$$(rA)^T = rA^T = rA.$$

Hence rA is in V. So V is closed under scalar multiplication.

Equivalently, you can justify that every symmetric matrix is of the form $\begin{bmatrix} a & b \\ b & c \end{bmatrix}$ for some $a, b, c \in \mathbb{R}$ (why?). Thus

$$V = \left\{ \begin{bmatrix} a & b \\ b & c \end{bmatrix} : a, b, c \in \mathbb{R} \right\} = \text{span} \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

is a subspace of $M_{2\times 2}$ since it is the span of the "vectors" $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, and $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$.

3. *Let $\mathbb{P}_n$ be the vector space of polynomials of degree at most n. Let W_1 be the set of all polynomials $p(t)$ in $\mathbb{P}_n$ which have a zero at $t = 1$ (that is, $p(1) = 0$), and let W_0 be the set of all polynomials $p(t)$ in $\mathbb{P}_n$ which have a zero at $t = 0$. Are these sets subspaces of the vector space of all polynomials?*

Both W_1 and W_0 are subspaces of $\mathbb{P}_n$. It is enough to check both are closed under linear combinations. So let $p(t), q(t)$ be two polynomials of the degree at most n such that $p(1) = q(1) = 0$. Then for c, d in $\mathbb{R}$, we want to check that $(cp + dq)(t)$ is also in W_1. So we have to check whether $(cp + dq)(1) = 0$. By the definition of addition and scalar multiplication for polynomials, we get that:

$$(cp + dq)(t) = cp(t) + dq(t)$$

Thus $(cp + dq)(1) = cp(1) + dq(1) = c0 + d0 = 0$. Thus W_1 is a subspace. Similarly one can show that W_0 is a subspace.

4. *In the class we introduced the vector space V of functions from $\mathbb{R}$ to $\mathbb{R}$. This problem will discuss a few aspect of this vector space.*

(a) Observe that $\sin(x), \cos(x)$ are in V and so are $2\sin(x)$ and $\sin(2x)$. Are $2\sin(x)$ and $\sin(2x)$ linear combinations of $\sin(x)$ and $\cos(x)$?

First, $2\sin(x) = 2\sin(x) + 0\cos(x)$ is obviously a linear combination of $\sin(x)$ and $\cos(x)$. Now, suppose $\sin(2x) = a\sin(x) + b\cos(x)$ for some $a, b \in \mathbb{R}$. Being an equality of functions, the equation must be true for all x in the common domain of the functions. For $x = 0$ we obtain $0 = 0 + b$ so that $b = 0$ yielding $\sin(2x) = a\sin(x)$. Next, setting $x = \pi/2$ yields $0 = a$ so that $\sin(2x) = 0\sin(x) + 0\cos(x) = 0$ the constant zero function. This contradiction shows that $\sin(2x)$ is not a linear combination of $\sin(x), \cos(x)$.

(b) Is the set of all continuous functions from $\mathbb{R}$ to $\mathbb{R}$ a subspace of V?

Yes. First, the 0 function is a continuous function. Let f_1 and f_2 be two continuous functions and c be a scalar. It is easy to see that both $f_1 + f_2$ and cf_1 are continuous.

(c) Is the set of all continuously differentiable functions from $\mathbb{R}$ to $\mathbb{R}$ a subspace of V?

Yes. First, the 0 function is a continuously differentiable function. Let f_1 and f_2 be two continuously differentiable functions and c be a scalar. Then both

$$\frac{d(f_1 + f_2)}{dx} = \frac{df_1}{dx} + \frac{df_2}{dx}$$

and

$$\frac{d(cf_1)}{dx} = c\frac{df_1}{dx}$$

exist and are continuous.

5. *Find an explicit description of the nullspaces of the following two matrices. You should write the nullspaces as spans of sets of vectors.*

$$A = \begin{bmatrix} 0 & 1 & 2 & 0 & 2 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}.$$

$\text{Nul}(A)$ is the set of all solutions to the homogeneous equation

$$\begin{bmatrix} 0 & 1 & 2 & 0 & 2 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

As A is already in reduced row echelon form, we identify x_2 and x_4 as pivot variables and x_1, x_3 and x_5 as the free variables. Thus we can write the general solution as

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} x_1 \\ -2x_3 - 2x_5 \\ x_3 \\ -3x_5 \\ x_5 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} 0 \\ -2 \\ 0 \\ -3 \\ 1 \end{bmatrix}$$

where $x_1, x_3, x_5 \in \mathbb{R}$ are free. We conclude that

$$\text{Nul}(A) = \text{span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -2 \\ 0 \\ -3 \\ 1 \end{bmatrix} \right\}.$$

Similarly, B is already in reduced row echelon form. We see that x_1 and x_4 are free variables, and $x_2 = -x_4$, $x_3 = -x_4$. Thus the general solution can be written as

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 0 \\ -1 \\ -1 \\ 1 \end{bmatrix}.$$

So we conclude that

$$\text{Nul}(A) = \text{span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ -1 \\ 1 \end{bmatrix} \right\}.$$

6. *Construct a matrix A such that*

$$\text{Nul}(A) = \text{span}\left(\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right).$$

Since the nullspace is a span of two independent vectors in $\mathbb{R}^3$, the matrix A must have exactly 3 columns and one pivot.
So let us assume $A = \begin{bmatrix} a & b & c \end{bmatrix}$. By the definition of the nullspace, for every vector $v \in \text{Nul}(A)$, we have $Av = 0$. In particular, we get

$$\begin{bmatrix} a & b & c \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 0 \qquad \begin{bmatrix} a & b & c \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = 0.$$

One solution of the above linear system is $A = \begin{bmatrix} a & b & c \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix}$.

7. *Let A, B be two matrices such that the product AB is defined. Is the nullspace of A contained in the nullspace of AB? Is the nullspace of B contained in the nullspace of AB? Explain why or why not.*

For every vector $v \in \text{Nul(B)}$, we have $Bv = 0$ and therefore

$$(AB)v = A(Bv) = A0 = 0,$$

which implies that v is also in $\text{Nul}(AB)$. Hence, we can conclude that $\text{Nul}(B) \subseteq \text{Nul}(AB)$.
However, the answer for the first question is no. Consider $A = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$, and $B = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$.
Then

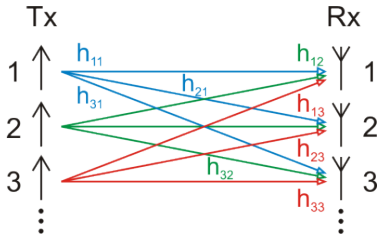
$$AB = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}.$$

It is not hard to get that

$$\text{Nul}(A) = \text{span}\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right), \qquad \text{Nul}(AB) = \text{span}\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right),$$

which do not contain each other.

8 (Wireless communication)**.** *The goal of this exercise is to better understand what a null space and a column space of a matrix represent in practical applications. In multiple-input and multiple-output (short: MIMO) systems, a transmitter sends multiple streams by multiple transmit antennas.*



Suppose there are n transmitters and m receivers. This can modeled using Linear Algebra:

$$\underbrace{\begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}}_{\text{received vector } \mathbf{y}} = \underbrace{\begin{bmatrix} h_{1,1} & \dots & h_{1,n} \\ \vdots & \ddots & \vdots \\ h_{m,1} & \dots & h_{m,n} \end{bmatrix}}_{\text{channel matrix } H} \underbrace{\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}}_{\text{transmitted vector } \mathbf{x}}.$$

So here the $\mathbf{x}$ *vector describes what the transmitter is sending out, and* $\mathbf{y}$ *is the vector describing what is received. The channel matrix* H *tells you how the signal* $\mathbf{x}$ *is transformed in to the vector* $\mathbf{y}$*.*

(a) In terms of what is received by the receiver, what does the first column of H represent? What does the i-th column represent?

The first column represents what the receivers receive from the first transmitter. Note that if the sender transmits the vector $\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ \dots \\ 0 \end{bmatrix}$, then the received vector $\mathbf{y}$ will be the first column of H. Similarly, the i-th column represents what the receivers receive from the i-th transmitter.

(b) If the signal $\mathbf{x}$ belongs to the nullspace of H, what signal $\mathbf{y}$ will be received?

The receivers will receive no signal, since $H\mathbf{x} = \mathbf{0}$.

(c) In a well designed system, what you want $\text{Nul}(H)$ to be?

We want that $\text{Nul}(H) = \{\mathbf{0}\}$. This guarantees that the receiver receive no signal if and only if no signal is transmitted.

(d) In terms of transmitted signals, what is $\text{Col}(H)$?

The set of signals that can be received. All the possible signals that can be received, are precisely the linear combinations of all the columns of H.

(e) If you take away one of the transmitters, how does H change and how does the column space of H change? What does that mean in terms of signals you can transmit?

Taking away a transmitter means deleting a column of H. Thus the column space potentially gets smaller. This again means that you will be able to receive fewer different signals. Note that the column space is will only change if the column is not all linear combination of the remaining columns.

9 (Redox reactions)**.** *Balance the following redox reaction:*

$$x_1\text{H}^+ + x_2\text{Cr}_2\text{O}_7^{2-} + x_3\text{C}_2\text{H}_5\text{OH} \leftrightarrow x_4\text{Cr}^{3+} + x_5\text{CO}_2 + x_6\text{H}_2\text{O}.$$

Describe the subspace of vectors $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix}$ *that balance this equation. (How can you make sure that the subspace is spanned by a single vector? Look at the electric charge!)*

One reads off the following equations

$$\begin{aligned} \text{H} &\Rightarrow x_1 + 6x_3 = 2x_6 \\ \text{Cr} &\Rightarrow 2x_2 = x_4 \\ \text{O} &\Rightarrow 7x_2 + x_3 = 2x_5 + x_6 \\ \text{C} &\Rightarrow 2x_3 = x_5 \end{aligned}$$

The augmented matrix is

$$\left[\begin{array}{cccccc|c} 1 & 0 & 6 & 0 & 0 & -2 & 0 \\ 0 & 2 & 0 & -1 & 0 & 0 & 0 \\ 0 & 7 & 1 & 0 & -2 & -1 & 0 \\ 0 & 0 & 2 & 0 & -1 & 0 & 0 \end{array}\right] \xrightarrow{R_2 \to \frac{1}{2}R_2} \left[\begin{array}{cccccc|c} 1 & 0 & 6 & 0 & 0 & -2 & 0 \\ 0 & 1 & 0 & -1/2 & 0 & 0 & 0 \\ 0 & 7 & 1 & 0 & -2 & -1 & 0 \\ 0 & 0 & 2 & 0 & -1 & 0 & 0 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 - 7R_2} \left[\begin{array}{cccccc|c} 1 & 0 & 6 & 0 & 0 & -2 & 0 \\ 0 & 1 & 0 & -1/2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 7/2 & -2 & -1 & 0 \\ 0 & 0 & 2 & 0 & -1 & 0 & 0 \end{array}\right]$$

$$\xrightarrow{R_4 \to R_4 - 2R_3} \left[\begin{array}{cccccc|c} 1 & 0 & 6 & 0 & 0 & -2 & 0 \\ 0 & 1 & 0 & -1/2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 7/2 & -2 & -1 & 0 \\ 0 & 0 & 0 & -7 & 3 & 2 & 0 \end{array}\right]$$

$$\xrightarrow{R_4 \to -R_4/7} \left[\begin{array}{cccccc|c} 1 & 0 & 6 & 0 & 0 & -2 & 0 \\ 0 & 1 & 0 & -1/2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 7/2 & -2 & -1 & 0 \\ 0 & 0 & 0 & 1 & -3/7 & -2/7 & 0 \end{array}\right]$$

$$\xrightarrow{R_3 \to -\frac{7}{2}R_3 - \frac{7}{2}R_4} \left[\begin{array}{cccccc|c} 1 & 0 & 6 & 0 & 0 & -2 & 0 \\ 0 & 1 & 0 & -1/2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1/2 & 0 & 0 \\ 0 & 0 & 0 & 1 & -3/7 & -2/7 & 0 \end{array}\right]$$

$$\xrightarrow{R_2 \to R_2 + \frac{1}{2}R_4} \left[\begin{array}{cccccc|c} 1 & 0 & 6 & 0 & 0 & -2 & 0 \\ 0 & 1 & 0 & 0 & -3/14 & -1/7 & 0 \\ 0 & 0 & 1 & 0 & -1/2 & 0 & 0 \\ 0 & 0 & 0 & 1 & -3/7 & -2/7 & 0 \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 - 6R_3} \left[\begin{array}{cccccc|c} 1 & 0 & 0 & 0 & 3 & -2 & 0 \\ 0 & 1 & 0 & 0 & -3/14 & -1/7 & 0 \\ 0 & 0 & 1 & 0 & -1/2 & 0 & 0 \\ 0 & 0 & 0 & 1 & -3/7 & -2/7 & 0 \end{array}\right]$$

We read off x_5 and x_6 as free variables and hence see that

$$x_1 = 2x_6 - 3x_5 \tag{1}$$

$$x_2 = \tfrac{3}{14}x_5 + \tfrac{1}{7}x_6 \tag{2}$$

$$x_3 = \tfrac{1}{2}x_5 \tag{3}$$

$$x_4 = \tfrac{3}{7}x_5 + \tfrac{2}{7}x_6. \tag{4}$$

Observe that

$$\dim \operatorname{Nul} \begin{bmatrix} 1 & 0 & 6 & 0 & 0 & -2 \\ 0 & 2 & 0 & -1 & 0 & 0 \\ 0 & 7 & 1 & 0 & -2 & -1 \\ 0 & 0 & 2 & 0 & -1 & 0 \end{bmatrix} = 2.$$

So without further information, we can not balance the chemical equation up to scalar multiple. However, because other electric charge of the molecule we get another equation:

$$x_1 - 2x_2 - 3x_4 = 0.$$

Thus

$$0 = (2x_6 - 3x_5) - 2(\tfrac{3}{14}x_5 + \tfrac{1}{7}x_6) - 3(\tfrac{3}{7}x_5 + \tfrac{2}{7}x_6).$$

Hence

$$x_5 = \tfrac{2}{11}x_6.$$

This gives us

$$x_1 = \tfrac{16}{11}x_6 \tag{5}$$

$$x_2 = \tfrac{2}{11}x_6 \tag{6}$$

$$x_3 = \tfrac{1}{11}x_6 \tag{7}$$

$$x_4 = \tfrac{4}{11}x_6. \tag{8}$$

Taking x_6 to be 11, we get

$$16\mathrm{H}^+ + 2\mathrm{Cr_2O_7^{2-}} + \mathrm{C_2H_5OH} \leftrightarrow 4\mathrm{Cr}^{3+} + 2\mathrm{CO_2} + 11\mathrm{H_2O}.$$